

Drill Drive Winch

Technical Datasheet - Generic Portal Reference

Document purpose: To provide a practical technical reference for theatre and entertainment use of drill-driven winch systems. This datasheet is intended to support inspection planning, portal documentation and site review. It does not replace manufacturer instructions, site-specific design information or competent-person assessment.

Equipment type	Drill Drive Winch
Portal category	Lifting Equipment
Typical classification	Powered / drill-assisted lifting or moving equipment
Document status	Generic reference template - edit for manufacturer and site details
Prepared for	Stage Electrics Mechanical Theatre Equipment Compliance Portal
Revision	Draft 1.0
Date	08 October 2026

Important limitation

This document is deliberately generic. The actual manufacturer, model, duty rating, approved drill type, drive adaptor, brake arrangement, rope type and installation method must be confirmed before the datasheet is used as equipment-specific technical information.

1. Equipment overview

A drill drive winch is a winch operated by an approved powered drill or similar drive tool. In a theatre or entertainment environment it may be used to raise, lower, tension or position production equipment such as bars, tracks, screens, scenery or lightweight suspended systems, depending on the design and manufacturer instructions.

- The winch, drive tool, drive adaptor, rope or wire rope, drum, guides, pulleys, terminations and supporting structure should be assessed as one complete system.
- The approved drive tool and operating settings are part of the safe operating arrangement.
- The system should only be used within its rated capacity, intended travel and manufacturer-defined duty limitations.
- Users should understand the risks of over-speed, overloading, poor control, incorrect drill selection and unintended movement.

- The arrangement should not be modified or used for a different purpose without competent review.

2. Equipment identification and technical details

Manufacturer	To be confirmed
Model / type	To be confirmed
Serial number / asset ID	To be confirmed
Location / installation reference	To be confirmed
Safe Working Load / Working Load Limit	To be confirmed from manufacturer or design information
Approved drive tool / drill type	To be confirmed - include model, voltage, clutch setting and speed range where specified
Drive adaptor / input connection	To be confirmed
Rope / wire rope / belt type	To be confirmed
Drum arrangement	To be confirmed - single layer / multi-layer / grooved / plain drum
Brake / holding method	To be confirmed
Limit or travel control	To be confirmed
Inspection interval	To be confirmed by written scheme, competent person or local inspection regime

3. Intended use

- Use only for the load type, travel distance, speed and mounting arrangement intended by the manufacturer or designer.
- Use only with the approved drive tool, drive adaptor and operating settings specified for the winch.
- Ensure the winch is securely mounted to a suitable structure before operation.
- Ensure the load path is known, including any rope redirects, pulleys, guide blocks and terminations.
- Maintain control of the drive tool at all times and avoid sudden starts, over-speed or shock loading.
- Do not lift people unless the complete system is specifically designed, rated and approved for that purpose.

4. Key limitations and hazards

Hazard / limitation	Why it matters	Control / check
Incorrect drill or setting	May create over-speed, excessive torque, poor stopping control or unintended movement.	Use only approved drive tools and settings. Confirm clutch and speed guidance.
Poor mounting	Can overload fixings or allow the winch to rotate or detach.	Check the fixing structure, bolts, anchors and alignment before use.
Rope damage or poor winding	Damaged or crossed rope can reduce capacity and cause sudden failure.	Inspect rope condition and winding before and during operation.
Brake or holding failure	The load may drift, descend or become uncontrolled.	Check brake / holding function and remove from service if unreliable.
Side loading or poor rope lead	Can damage the drum, rope, guides or fixing points.	Keep rope lead aligned and use suitable pulleys or guides.
Unverified modification	Changes can invalidate rating or manufacturer assumptions.	Do not modify without competent assessment and documentation.

5. Inspection and maintenance guidance

The following checks should be adapted to the manufacturer instructions, written scheme of examination and local inspection regime. Items with defects, missing rating information or uncertain compatibility should be withdrawn from service until assessed by a competent person.

Area	Typical checks	Status	Notes
Identification and rating	SWL/WLL, serial number, asset ID, intended use and manufacturer markings are present and legible.	☐ OK ☐ Defect	
Winch body and mount	Frame, casing, drum, fixing bolts, mounting plate and support structure are secure and undamaged.	☐ OK ☐ Defect	
Drive tool and adaptor	Approved drill/tool, drive bit/adaptor, coupling, torque setting and speed range are suitable and undamaged.	☐ OK ☐ Defect	

Area	Typical checks	Status	Notes
Rope / wire rope / belt	Check for wear, broken wires, kinks, crushing, abrasion, corrosion, contamination or poor winding.	☐ OK ☐ Defect	
Terminations and anchors	Sockets, clamps, thimbles, wedge fittings, anchor points and rope dead ends are correctly fitted.	☐ OK ☐ Defect	
Pulleys, guides and sheaves	Check free rotation, alignment, bearing condition, groove wear, guards and secure fixing.	☐ OK ☐ Defect	
Brake / holding system	Load can be stopped and held safely without drift, creep or sudden release.	☐ OK ☐ Defect	
Operation	Movement is smooth, controlled and free from abnormal noise, vibration, slipping or snatching.	☐ OK ☐ Defect	
Documentation	Inspection reports, maintenance records, manufacturer instructions and site procedures are available.	☐ OK ☐ Defect	

6. Pre-use checks

- Confirm the winch and support point are suitable for the load and direction of pull.
- Confirm the approved drill or drive tool is available and correctly set.
- Check that the rope, wire rope or belt is correctly seated on the drum and not trapped, crossed or damaged.
- Check that the load is free to move and that the travel path is clear.
- Check that all persons are clear of the load path and pinch points.
- Operate slowly at first and stop immediately if abnormal movement, noise, slipping or poor control is observed.

7. Regulatory and good-practice context

Framework	Relevance to drill drive winch systems
LOLER 1998	Likely relevant where the winch is used for lifting or lowering loads at work. The equipment and lifting operation should be planned, suitable, marked and thoroughly examined where required.
PUWER 1998	Applies as work equipment. The winch, drive tool and associated parts must be suitable, maintained, inspected where deterioration could create risk, and used by trained people.
BS EN 17206	Relevant where the winch forms part of entertainment machinery or a stage/production lifting system. Control of movement, load path integrity and documentation should be considered.
ABTT Technical Standards	Supports theatre-industry good practice for installation, inspection, maintenance and competent operation in places of entertainment.

Competent-person note

The competent person should determine the inspection and thorough examination requirements for the specific installation, taking account of manufacturer guidance, duty, environment, frequency of use and consequences of failure.

8. Information to retain with this datasheet

- Manufacturer instructions and Declaration of Conformity where applicable.
- Design or installation information confirming support points and load path.
- Proof load / test information where applicable and appropriate.
- Thorough examination reports and inspection records.
- Maintenance records, defect reports and repair history.
- Approved drive tool details and operating settings.
- Site-specific risk assessment and safe operating procedure.

9. Portal upload details

Recommended PDF filename	drill_drive_winch_technical_datasheet.pdf
Recommended GitHub path	documents/drill_drive_winch_technical_datasheet.pdf
Portal document title	Drill Drive Winch Technical Datasheet
Related equipment ID	drill_drive_winch

10. Revision notes

Revision	Date	Change	By
1.0	08 Oct 2026	Initial generic portal datasheet.	Stage Electrics / ChatGPT support

Disclaimer: This datasheet is a guidance aid only. It must be reviewed and completed using the actual manufacturer documentation, installation records and competent-person findings before being relied on as equipment-specific technical data.
